24/sc/MA/028

UNIVERSITY OF UYO
DEPARTMENT OF MATHEMATICS
2025/2026 FIRST SEMESTER EXAMINATION

COURSE: MTH216: SCIENTIFIC COMPUTING I
DATE SCHEDULED: THURSDAY 12th APRIL, 2026 ★ TIME ALLOWED: TWO HOURS (2 HRS)

Instructions: *Answer question one (1) and attempt any other three (3) questions only.*

1. (a) *What is Maple?* — 2 *marks*
   (ii) List three mathematical and relational opreators in Maple. — 3 *marks*
   (iii) List three constants in Maple . — 3 *marks*
   (iv) State three functions in Maple. — 3 *marks*
   (v) State three control flow structures in Maple. — 3 *marks*

   (b) (i) What is Matlab? — 2 *marks*
   (ii) List three mathematical and relational opreators in Matlab. — 3 *marks*
   (iii) List three constants in Matlab. — 3 *marks*
   (iv) State three functions in Matlab. — 3 *marks*
   (v) State three control flow structures in Matlab. — 3 *marks*

2. Define any function of your choice with domain, plot two graphs using: (a) Maple and (b) Matlab. — 7 *marks*

3. (a) (i) Write a program using *solve command* to solve the equation $x^2 + x + 1 = 0$ in Maple. — $3\frac{1}{2}$ *marks*
   (ii) Write a program using *solve command* to solve the equation $x^2 - 4y^2 = 1, x - 8y^3 = 3$ in Maple. — $3\frac{1}{2}$ *marks*

   (b) Write a program using *solve command* to solve the equation $x^2 + x + 1 = 0$ in Matlab. — $3\frac{1}{2}$ *marks*
   (ii) Write a program using *solve command* to solve the equation $x^2 - 4y^2 = 1, x - 8y^3 = 3$ in Matlab. — $3\frac{1}{2}$ *marks*

4. (a) Defining Simple Procedures in Maple to solve quadratic equation. — $3\frac{1}{2}$ *marks*
   (b) Write a function program to solve quadratic equation in Matlab. — $3\frac{1}{2}$ *marks*

5. (a) Write for loop program in Maple to compute $\sum_{i=1}^{25} i$. — $3\frac{1}{2}$ *marks*

   (b) Write for loop program in Matlab to compute $\sum_{i=1}^{25} i$ — $3\frac{1}{2}$ *marks*

14
14
21
49